

Andrew Russell

CRYPTOGRAPHER · EX-GOOGLE · EMERGING-MARKETS OPERATOR — AI AGENT SAFETY

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Summary

Cryptography researcher and software engineer turned emerging-markets product operator, now working on AI agent safety. Published at CRYPTO 2018; shipped security software at Google and Microsoft; four years running data, product, and P&L for an agri-fintech across Kenya and Zambia. **Currently on sabbatical (returning Sept 2026):** designing truly autonomous AI agents with formally-analyzed safety, security, and liveness properties (see Costanza, 2026) and motorbiking from Nairobi to Namibia. Most interested in product roles at the intersection of AI research, safety, and real-world deployment.

Work Experience

Sabbatical (returning to full-time Sep 2026)

Amsterdam / Nairobi

Independent AI safety research & travel

Jan 2026 – Present

- Designed, built, and shipped **Costanza**, a fully autonomous AI agent, complete with a formal security and liveness model. See Projects below.
- Solo motorbike expedition from Kenya to Namibia, May–Aug 2026.

Apollo Agriculture

Netherlands / Kenya / Zambia

Senior Product Manager

May 2024 – Dec 2025

- Authored two company-wide retrospectives following major customer contraction, combining market research, customer/staff feedback, and internal operational analysis; findings drove pivots in business model and multi-year growth strategy.
- Appointed by C-suite to lead Zambia's \$10M revenue business unit; accountable for commercial performance, product, and operations across sales, credit, and engineering.
- Built experimental products, including a wholesaling initiative to reduce balance sheet risk from multi-million-dollar fertilizer holdings.

Software Engineer → Tech Lead → Head of Data

Apr 2021 – Mar 2024

- Built and scaled Apollo's core data infrastructure. Led migration of legacy analytics to modern stack, improving query speeds by 97% and cutting costs by 60%. Wrote 100,000+ lines of production SQL and Python.
- Led cross-functional analytics, enabling multi-country operations and supporting record 200% annual growth.
- Promoted to Head of Data; managed a team of 8 engineers and analysts across Europe and Africa.

Google

San Francisco, CA

Software Engineer

Aug 2019 – Mar 2021

- Built a system monitoring petabytes of user data for daily privacy compliance across Google's product ecosystem.

Software Engineer, PhD Intern

Summer 2018

- Designed a threat model for a privacy-preserving version of Gmail using secure execution environments; ported part of Gmail's email-processing backend into the system.

Microsoft

Redmond, WA

Software Engineer

Aug 2016 – Aug 2017

- Worked on Windows Security, particularly secure sign-in/sign-out flows for Windows 10 (Windows Hello). One-year role between undergrad and Ph.D. program.

Projects

Costanza: The Human Fund (2026)

thehumanfund.ai · github

Fully autonomous, indestructible AI agent, built as a proof-of-concept warning about such extrajudicial agents. Designed with constrained alignment (provably cannot do harm even under prompt injection), hardware attestation (Intel TDX), and formal security + liveness model with game-theoretic analysis. [Whitepaper on GitHub](#).

Publications

Risky Traitor Tracing and New Negative Differential Privacy Results

with Rishab Goyal, Venkata Koppula, and Brent Waters

CRYPTO 2018 (top-tier cryptography venue)

Education

The University of Texas at Austin

M.S. in Computer Science & Ph.D. coursework

Cryptography (obfuscation, homomorphic/functional encryption). Advised by Brent Waters.

Austin, TX

2017 – 2019

Pomona College

B.A. in Mathematics and Computer Science

Thesis: Elliptic Curve Pairings in Cryptography. Summer research fellowships at Stanford (2014) and the University of Bristol (2015).

Claremont, CA

2012 – 2016

Fun Stuff

Ultramarathoner, Division I club athlete, and 2x Dutch national champion in ultimate frisbee. Eagle Scout, thru-hiker & backpacker across 25+ countries. Violinist for 15+ years; former member of the Phoenix Youth Symphony. Founded & managed a world-champion e-sports team in high school. SCUBA certified. Adventure motorcyclist with 10,000+ km on the odometer.